

AYUSH KUMAR RAY

Software Engineer | Java | C++ | DSA | Software Architecture | IoT

Parul University, Vadodara, Gujarat

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Open to relocation and full-time/in-office roles

Professional Summary

Software Engineer | 250+ DSA problems solved (Java/C++) | 20+ REST APIs built | Distributed Systems & IoT

Built fault-tolerant, real-time backend systems (FastAPI, PostgreSQL/PostGIS, Redis, Firebase) and an **IoT-based accident-detection pipeline** (Arduino/Raspberry Pi, MQTT) for an emergency-response platform. Strong foundation in **Object-Oriented Design, Software Architecture, Digital Systems, and Operating Systems**, with hands-on experience translating requirements into modular, production-ready code.

Education

Parul University

2023 – 2027

Bachelor of Technology (B.Tech) in Computer Science and Engineering

Vadodara, Gujarat

• **CGPA: 7.80/10**

• **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Digital Systems, Software Engineering, Distributed Systems, Database Management Systems, System Design Fundamentals

Technical Skills

Core Match: IoT (Arduino, Raspberry Pi, MQTT, Sensor Integration) • Digital Systems (Truth Tables, State Machines) • Computer Architecture • Operating Systems

Languages: Java, C++, Python, JavaScript, TypeScript, SQL

Systems & Architecture: OOD, Software Architecture, Distributed Systems, Fault-Tolerant Design, SDLC, Design Patterns, Clean Architecture

Backend: RESTful APIs, WebSockets, Event-Driven Architecture, FastAPI, SQLAlchemy, Pydantic, JWT Auth, Concurrent/Async Processing

Testing & QA: Test Plan Development, System Testing, Unit & Integration Testing (pytest), Load Testing, Code Review

Data & Cloud: PostgreSQL/PostGIS, MySQL, Redis, AWS (EC2, S3, RDS), Docker, CI/CD

Tools: Git, GitHub, Visual Studio IDE, MS Office

Web Technologies: React.js, Next.js, HTML5, CSS3, Tailwind CSS

Projects

Emergency Response & Hospital Coordination Platform | FastAPI, PostgreSQL/PostGIS, Redis, WebSockets, Firebase, IoT | **2026**

- **Enabled automatic emergency dispatch, no manual reporting delay** — built an IoT accident-detection system using Arduino/Raspberry Pi airbag-crash sensors, streaming data to the backend via MQTT/HTTP.
- **Sustained correct hospital-matching during partial outages** — architected a fault-tolerant backend applying software-architecture and OS principles, achieving zero incorrect routing in failure simulations.
- **Prevented regressions across 20+ decoupled API endpoints** — wrote test plans and unit/integration tests (pytest), applying OOD to isolate failures by module.
- **Cut geospatial query latency and enabled real-time sync** — implemented a WebSocket + Firebase layer over TCP/IP/HTTP with PostgreSQL/PostGIS indexing.

LifePulse AI – Smart Blood Donation Network | FastAPI, PostgreSQL/PostGIS, Redis, WebSockets, Firebase | [Live](#) | [GitHub](#) | **2026**

- **Prioritized the nearest compatible donors in real time** — designed a geospatial distance and blood-type scoring algorithm for emergency donor-ranking.
- **Improved donor-matching scalability** — applied database design and indexing best practices with normalized schemas in PostgreSQL/PostGIS.
- **Increased notification delivery reliability** — implemented WebSocket and Firebase Cloud Messaging pipelines with fallback escalation across donor pools.
- **Improved maintainability, validated stability under load** — applied object-oriented design and clean architecture, tuning Redis caching and indexing for confirmed performance thresholds.

Algorithms & Problem Solving

- **250+ DSA problems solved** in Java/C++ across LeetCode, CodeChef, HackerRank, and Smart Interviews – Trees, Graphs, DP, Greedy, Hashing, Heaps, Recursion, Backtracking, Binary Search, Complexity Analysis
- **Certification:** Smart Coder (Bronze) – Smart Interviews

Certifications

AWS Academy Graduate – Cloud Foundations

Artificial Intelligence Fundamentals

ServiceNow Certified System Administrator (CSA)

ServiceNow Certified Application Developer (CAD)

Building RAG Agents with LLMs

Building Agentic AI Applications with LLMs

Amazon Web Services (AWS)

IBM SkillsBuild

ServiceNow

ServiceNow

Nvidia

Nvidia